

Resampling-Based Model Selection for PE SDFs

How to empirically measure the risk-adjusted performance of non-traded cash flows

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1 Introduction

Which stochastic discount factor (SDF) model gives the most credible risk adjustment for private-equity cash flows? This paper argues that the answer depends critically on a question that existing work treats as settled: *how should the pricing moments be constructed?*

The core problem: moment construction is consequential. Private-equity samples are small, lumpy, and organized by vintage year. An SDF model can look good or bad depending on how funds are grouped into pricing moments. Existing approaches commit to one grouping—typically single funds (e.g., Ang et al. [2018]) or vintage-year portfolios (e.g., Driessen et al. [2012])—and treat it as fixed. But the choice of grouping is itself consequential for estimation, and there is no a priori reason to prefer one admissible grouping over another.

We propose to make the grouping itself part of the empirical design. We repeatedly draw random within-vintage fund partitions, estimate the SDF on partition-induced moments, and evaluate pricing errors on held-out moments. The preferred SDF model is the one with the lowest average holdout pricing loss across many admissible partitions. We call this *resampling-based model selection*: moment construction becomes a testable design choice rather than a fixed input.

Two additional difficulties arise within this framework and must be handled by modular design choices:

Unseasoned vintages create a dilemma. Recent vintage years carry valuable pricing information—they reflect current market conditions and expand the factor-path coverage of the sample—but including them naively destabilizes (or biases) estimation. The standard approach treats the latest reported NAV as a terminal cash flow and discounts it over the fund’s entire remaining horizon. This creates two problems. First, it concentrates a large share of the fund’s discounted value at a single calendar date—the dataset endpoint—so that the estimator’s fit becomes disproportionately sensitive to the macro state at that date rather than to the fund’s lifetime factor exposure. Second, the terminal NAV is an appraisal, not a realized payoff; its co-movement with public indices may differ systematically from that of actual divestitures. Researchers therefore face a dilemma: exclude recent vintages and waste data, or include them and accept fragile estimates driven by one macro-state realization. Rather than commit to one approach, our framework accommodates and empirically compares three competing strategies for unseasoned vintages (Section 6).

Exploding-alpha degeneracy in finite samples. NPV-style SDF estimation suffers from a finite-sample pathology first documented by Driessen et al. [2012]. In the standard multiplicative SDF parameterization, cash flows are discounted by a cumulative factor that includes $(1 + \alpha)^{-T}$. When α is large and positive, this factor drives all discount factors toward zero for large T , making every fund’s pricing error approximately zero regardless of the factor loadings β . The result is a ridge of spurious near-minima along the alpha axis of the objective function. The problem is most severe when fund horizons are long and early cash flows are small relative to later ones, which is precisely the cash-flow shape of a typical PE fund. The resampling framework controls this pathology because the validation fold penalizes degenerate alpha values: extreme estimates that overfit estimation folds produce large holdout pricing errors (Section 7).

Summary of the approach. The method has three interrelated goals:

- **Model selection:** rank SDF models by out-of-sample pricing performance across many resampled

moment constructions.

- **Design comparison:** compare whether within-vintage fund resampling or traditional vintage-year portfolio construction yields more stable and predictive pricing moments.
- **Efficient use of the vintage panel:** include all available vintages—seasoned and unseasoned—through flexible, smart resampling designs tailored to different approaches for handling unseasoned data.

The paper is closest in spirit to Korteweg and Nagel [2016] because the estimator works with aggregated pricing moments, and to Driessen et al. [2012] because the moments price private cash flows directly. The new contribution is a general resampling-based model selection framework that (i) treats moment construction as a testable design choice, (ii) nests competing approaches to unseasoned vintages as special cases and ranks them empirically, and (iii) can accommodate time-varying factor specifications through vintage-stratified estimation (Section 9). Brown et al. [2023] also use NAVs to estimate the risk and return of PE funds. Ghysels et al. [2025] construct high-frequency PE returns from NAV-based state-space models and use them for factor analysis and spanning tests.

2 Guiding Example

Suppose a candidate SDF has three parameters: an alpha term and two factor loadings. Then we need at least three pricing moments to estimate it. A natural resampling design is therefore:

- build three estimation folds,
- build one validation fold,
- estimate the SDF on the three estimation moments,
- score the fitted SDF on the validation moment (calculate out-of-sample pricing error).

In stylized notation, the example can be summarized with one simple three-parameter SDF and one resampled moment per fold. Let the parameter vector be

$$\theta = (\alpha, \beta_1, \beta_2)',$$

and let the one-period discount factor be

$$M_{t-1,t}(\theta) = \frac{1}{1 + \alpha + \beta_1 f_{1,t} + \beta_2 f_{2,t}}, \quad \Psi_{0,t}(\theta) = \prod_{s=1}^t M_{s-1,s}(\theta),$$

where $f_{1,t}$ and $f_{2,t}$ are two benchmark factor realizations. For sampled split n , let $C_{n,v,r}$ be the funds from vintage v assigned to fold $r \in \{1, 2, 3, 4\}$, with $r = 1, 2, 3$ used for estimation and $r = 4$ used for validation. Write $CF_{t,i} = D_{t,i} - X_{t,i}$, and partition active vintages into seasoned and unseasoned sets, $\mathcal{V} = \mathcal{V}^S \cup \mathcal{V}^U$. The vintage-level pricing restriction is then

$$\epsilon_{n,v,r}(\theta) = \begin{cases} \epsilon_{n,v,r}^S(\theta) = \frac{1}{|C_{n,v,r}|} \sum_{i \in C_{n,v,r}} \frac{1}{K_i} \sum_{t=1}^{T_i} \Psi_{0,t}(\theta) CF_{t,i}, & v \in \mathcal{V}^S, \\ \epsilon_{n,v,r}^U(\theta) = \frac{1}{|C_{n,v,r}|} \sum_{i \in C_{n,v,r}} \Lambda_i \frac{1}{T_i} \sum_{t=1}^{T_i} \frac{\eta_{t,i}(\theta)}{s_{t-1,i}}, & v \in \mathcal{V}^U, \end{cases}$$

where the unseasoned one-period Euler error is

$$\eta_{t,i}(\theta) = M_{t-1,t}(\theta)(CF_{t,i} + \text{NAV}_{t,i}) - \text{NAV}_{t-1,i}.$$

Here $s_{t-1,i} = \max\{\text{NAV}_{t-1,i}, \nu K_i\}$ converts each quarterly Euler error into a relative error per dollar at risk, while Λ_i is a predetermined duration-matching normalizer based on fund age and strategy that maps the fund-average relative Euler error into an NPV-like quantity per committed dollar. The resampled fold moment is the cross-vintage average

$$m_{n,r}(\theta) = \frac{1}{|\mathcal{V}|} \sum_{v \in \mathcal{V}} \epsilon_{n,v,r}(\theta).$$

Because θ has three components, the three estimation folds deliver three stylized moment conditions,

$$m_{n,1}(\theta) = 0, \quad m_{n,2}(\theta) = 0, \quad m_{n,3}(\theta) = 0,$$

or equivalently

$$\hat{\theta}_n = \arg \min_{\theta} \sum_{r=1}^3 m_{n,r}(\theta)^2.$$

The held-out fourth fold then gives the validation score

$$S_n(\hat{\theta}_n) = m_{n,4}(\hat{\theta}_n)^2.$$

Our resampling-based approach repeats this exercise many times: for each draw $n = 1, \dots, N$, we redraw the partition and thereby obtain a new quartet of fold moments $\{m_{n,1}, m_{n,2}, m_{n,3}, m_{n,4}\}$.

Figures 1 and 2 show the two baseline ways to build these folds. In both figures, each dot is one fund. Vintages 2005–2010 lie above the approach boundary and use cash-flow discounting, while vintages 2011–2012 lie below it and use the normalized Euler approach.

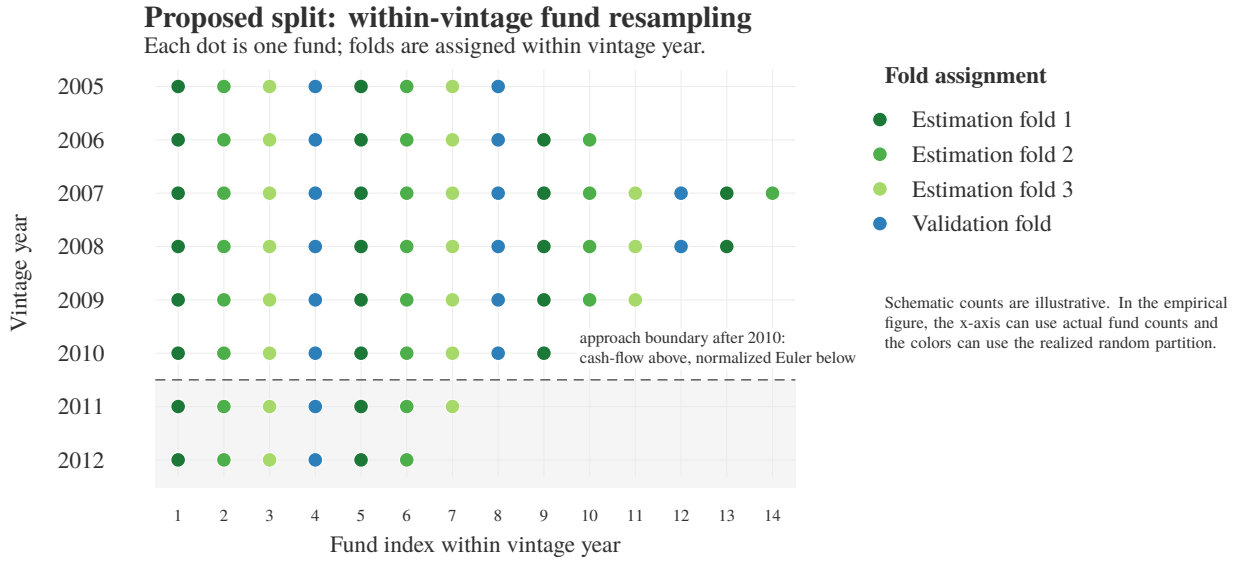


Figure 1: Proposed within-vintage fund resampling scheme. Each active vintage is split into three estimation cells and one validation cell. Fold 1 contains the dark-green funds from every vintage, fold 2 contains the medium-green funds from every vintage, fold 3 contains the light-green funds from every vintage, and the validation fold contains the blue funds from every vintage.



Figure 2: Traditional shuffled vintage-year portfolio scheme. Funds are first aggregated into whole vintage-year portfolios, and whole vintages are then assigned to folds. Here, 2005 and 2009 enter estimation fold 1, 2006 and 2010 enter estimation fold 2, 2007 and 2011 enter estimation fold 3, and 2008 and 2012 form the validation fold.

3 Framework and Notation

Let $v = 1, \dots, V$ index active vintage years, and let $\mathcal{I}_v$ be the set of funds in vintage v . Let M denote a candidate SDF model with parameter vector $\theta \in \Theta_M \subset \mathbb{R}^{p(M)}$.

We use P estimation folds and h holdout folds. For the guiding three-parameter example, $P = 3$ and $h = 1$. When comparing models with different numbers of parameters, a clean implementation is to use a common number of estimation folds $P = p_{\max}$, where $p_{\max} = \max_M p(M)$.

3.1 Within-vintage resampling

In the proposed design, every fold contains funds from every active vintage. This keeps the vintage-year support fixed across estimation and validation moments. The folds differ because they contain different funds, not because they represent different macro or vintage states. This is the cleanest design for asking whether an SDF prices PE cash flows robustly across fund-level resamples.

For sampled split $n = 1, \dots, N$, split the funds within each vintage into $P + h$ cells:

$$\mathcal{P}_{n,v} = \{C_{n,v,1}, \dots, C_{n,v,P+h}\}, \quad C_{n,v,r} \subset \mathcal{I}_v.$$

The cells are disjoint and cover the funds in vintage v . The r th cross-vintage fold is

$$F_{n,r} = \{C_{n,1,r}, C_{n,2,r}, \dots, C_{n,V,r}\}.$$

Thus fold r contains one fund sub-sample from every active vintage.

At the single-fund level, the number of admissible assignments is usually too large to enumerate. We therefore draw Monte Carlo samples from a partition rule Π :

$$\mathcal{P}_n \sim \Pi(\cdot \mid P, h, \mathcal{I}_1, \dots, \mathcal{I}_V).$$

For now, Π should be simple and transparent: random assignment within each vintage, subject to no empty cells and reasonable balance in fund counts or economic size.

3.2 Vintage-year portfolio benchmark

In the traditional design, the primitive object is a vintage-year portfolio. A fold contains whole vintages, not fund sub-samples from all vintages. This is easy to explain and resembles standard PE portfolio construction. But the validation moment now differs from the estimation moments in vintage composition. A failure on the validation fold may therefore reflect poor SDF pricing, or it may reflect extrapolation across vintage states.

The notation is the same at the fold level, but the economic object differs: $F_{n,r}$ now contains whole vintage-year portfolios rather than within-vintage fund cells.

3.3 Design comparison

The first version of the paper should focus on this contrast. If within-vintage resampling delivers more stable estimates and lower holdout pricing errors, the evidence favors building all-vintage subportfolios before estimating the SDF. If shuffled vintage-year portfolios perform similarly, the simpler vintage-year construction may be sufficient.

4 Pricing Moments

For a fold $F_{n,r}$, let $\epsilon_{n,v,r}(\theta; M, \eta)$ be the pricing error from the vintage component of that fold. The design choice η collects choices such as the alpha definition, loss function, and cash-flow weighting.

The aggregated fold moment is

$$m_{n,r}(\theta; M, \eta) = \sum_{v \in A_{n,r}} \omega_{n,v,r} \epsilon_{n,v,r}(\theta; M, \eta),$$

where $A_{n,r}$ is the set of vintages represented in fold r , and $\omega_{n,v,r}$ are vintage or value weights. Under within-vintage resampling, $A_{n,r}$ is the full active vintage set for every fold. Under shuffled vintage-year portfolios, $A_{n,r}$ is only the set of whole vintages assigned to fold r .

This distinction is the core empirical design choice. The proposed method creates comparable fold moments that all span the same vintage domain. The benchmark creates moments from different vintage domains.

4.1 Fund weights and vintage weights

The fund-level scaling and vintage weight are design choices that determine the economic content of the fold moment. For seasoned funds, the baseline contribution is

$$g_i^S(\theta; M, \eta) = \frac{1}{K_i} \sum_{t=0}^{T_i} \Psi_{0,t}(\theta; M, \eta) CF_{t,i},$$

so each seasoned fund contributes a *relative* NPV error per dollar committed. For unseasoned funds, the analogous baseline contribution is the normalized Euler error $g_i^U(\theta; M, \eta)$ defined in Section 6.3: a fund-average one-period Euler error scaled by lagged capital at risk and rescaled by a deterministic duration factor. At the cell level, we average over the funds in the cell. At the vintage level, the baseline sets $\omega_{n,v,r} = 1/|A_{n,r}|$ (equal vintage weights), so that each vintage contributes equally to the fold moment regardless of the number of funds or aggregate capital it contains.

Under this hybrid three-layer normalization (fund $\rightarrow$ cell $\rightarrow$ vintage), the fold moment becomes

$$m_{n,r}(\theta; M, \eta) = \frac{1}{|A_{n,r}|} \left[\sum_{v \in \mathcal{V}^S \cap A_{n,r}} \frac{1}{|C_{n,v,r}|} \sum_{i \in C_{n,v,r}} g_i^S(\theta; M, \eta) + \sum_{v \in \mathcal{V}^U \cap A_{n,r}} \frac{1}{|C_{n,v,r}|} \sum_{i \in C_{n,v,r}} g_i^U(\theta; M, \eta) \right].$$

This is an average, across vintages, of the average relative pricing error per fund. For seasoned vintages the object is a relative NPV error; for unseasoned vintages it is a duration-matched relative Euler error expressed in comparable per-committed-dollar units. It is therefore comparable across: (i) vintages with different numbers of funds, (ii) vintages with different aggregate capital, (iii) seasoned vintages with cash-flow errors and unseasoned vintages with normalized Euler errors (Section 6.3), and (iv) estimation folds and validation folds.

4.2 Choice of pricing moment

The pricing error in the loss L is a design choice, not a primitive. Two canonical families appear in the private-capital SDF literature: the *dollar NPV* moment of Driessen et al. [2012] and the *PME-ratio* moment of Kaplan and Schoar [2005], Driessen et al. [2009], and Ang et al. [2018]. We treat the NPV moment as the baseline and the PME-ratio as a robustness check.

NPV moment (baseline). For seasoned funds, aggregate discounted cash flows into one cumulative pricing error per dollar committed. For unseasoned funds, use the normalized Euler fund moment $g_i^U(\theta; M, \eta)$ from Section 6.3. The empirical baseline is therefore hybrid: cumulative NPV for $\mathcal{V}^S$, normalized Euler for $\mathcal{V}^U$. The NPV moment is linear in cash flows, additive across funds, and compatible with the normalized Euler NAV restriction for unseasoned vintages. Both blocks are expressed as per-committed-dollar pricing errors, and neither requires ratio denominators that can approach zero.

PME-ratio moment (robustness check). Define the risk-adjusted PME of fund i as the ratio of discounted distributions to discounted contributions. This is scale-invariant and has natural practitioner interpretability, but it is nonlinear in θ , numerically fragile when the denominator is small, and has no clean short-horizon analogue for unseasoned vintages—integrating $\mathcal{V}^U$ into a PME framework typically requires treating the latest NAV as a terminal distribution and discounting it over the fund’s entire remaining horizon, reintroducing the long-horizon instability the normalized Euler block is designed to avoid. We therefore restrict the PME robustness check to seasoned vintages $\mathcal{V}^S$. Detailed pros and cons of each moment family are given in Appendix A.

Robustness diagnostic. Stability of the preferred model-design pair (M, η, Π) across the hybrid NPV/Euler baseline, the raw dollar NPV, and the pooled-PME moment (restricted to $\mathcal{V}^S$) is the appropriate robustness diagnostic: agreement suggests genuine model fit, while disagreement would flag moment-induced identification.

5 Estimation and Holdout Scoring

For each sampled split n , let E_n be the set of estimation folds and H_n the validation fold. The split-specific estimator solves

$$\hat{\theta}_n(M, \eta, \Pi) = \arg \min_{\theta \in \Theta_M} \frac{1}{P} \sum_{j \in E_n} L(m_{n,j}(\theta; M, \eta)).$$

The validation score is the pricing loss on the held-out fold:

$$S_n(M, \eta, \Pi) = L(m_{n,H_n}(\hat{\theta}_n(M, \eta, \Pi); M, \eta)).$$

Aggregate this score across many sampled assignments:

$$\text{Score}_N(M, \eta, \Pi) = \frac{1}{N} \sum_{n=1}^N S_n(M, \eta, \Pi).$$

The preferred model-design pair (M, η, Π) is the one with the lowest average holdout pricing loss. Parameter stability is still informative, but the primary ranking criterion should be predictive pricing performance: does the fitted SDF price cash flows that were not used in estimation?

5.1 Why validation is needed

If a model has p parameters and we estimate it on exactly p moments, the in-sample fit can look deceptively good. This is especially dangerous in small PE samples because exact identification can hide instability. The validation fold creates a more demanding question: after fitting the model on P moments, does it also price a held-out PE cash-flow portfolio?

The main empirical specifications should be:

- **Main comparison:** $P = p_{\max}$, $h = 1$, so all candidate models are compared on a common fold architecture.
- **Exact-identified diagnostic:** $P = p(M)$, $h = 0$, to study feasibility and parameter instability.
- **Model-specific validation:** $P = p(M)$, $h = 1$, to compare exact-identified estimation plus one holdout fold.
- **Scheme comparison:** run the same model list under within-vintage resampling and shuffled vintage-year portfolios.

6 Approaches to Unseasoned Vintages

Partition the active vintages into seasoned and unseasoned sets,

$$\mathcal{V} = \mathcal{V}^S \cup \mathcal{V}^U.$$

A vintage v is *unseasoned* ($v \in \mathcal{V}^U$) if its aggregate residual NAV exceeds a fraction δ of total committed capital; otherwise it is *seasoned* ($v \in \mathcal{V}^S$). In practice, $\delta \in [0.10, 0.20]$ is a sensible range. For seasoned vintages, the pricing restriction is the standard cumulative cash-flow error as defined in Section 4.

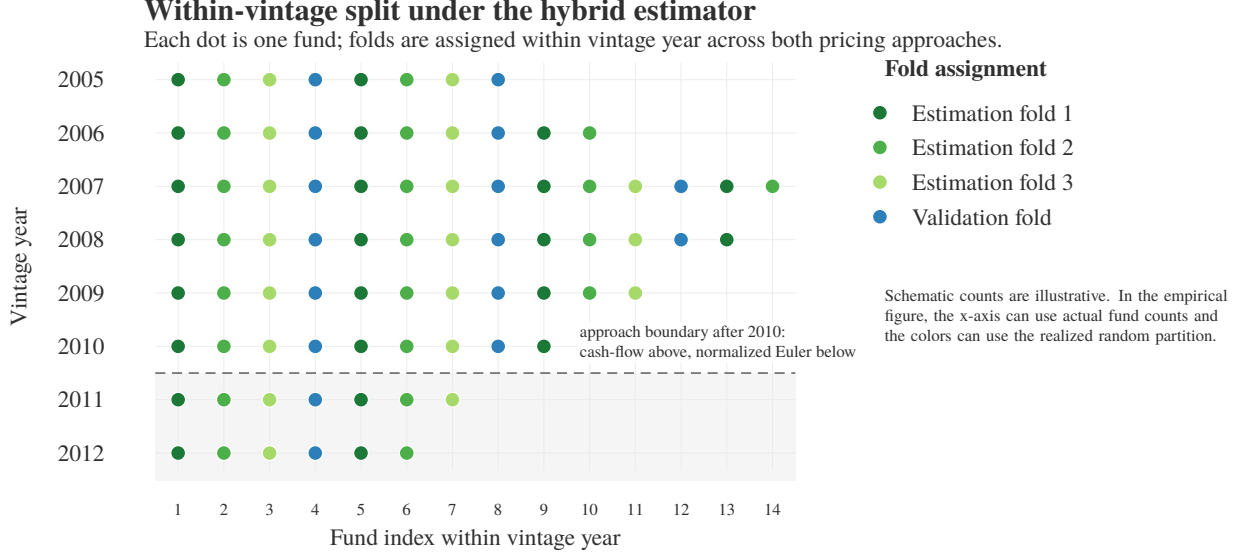


Figure 3: Within-vintage fold assignment under the hybrid cash-flow / normalized-Euler estimator. Unseasoned vintages 2011–2012 ($\mathcal{V}^U$, grey band) are included in estimation and validation like the seasoned vintages ($\mathcal{V}^S$), but contribute the normalized one-period Euler NAV error ϵ^U in place of the cumulative cash-flow error ϵ^S .

The central question is how to include unseasoned vintages—or whether to include them at all. Rather than commit to one approach a priori, our framework accommodates three competing strategies and lets the resampling-based holdout loss rank them empirically.

6.1 Approach A: NAV-adjusted cash-flow SDF

The most direct strategy adjusts the terminal NAV before applying the standard cumulative cash-flow SDF. For seasoned vintages, the pricing restriction is unchanged. For unseasoned vintages, the latest reported NAV is multiplied by an empirical discount factor d_i before being treated as a terminal distribution:

$$\tilde{D}_{T,i} = D_{T,i} + d_i \cdot \text{NAV}_{T,i},$$

where $d_i \in (0, 1]$ is the NAV discount. The fund then enters the standard CF-SDF pricing restriction using $\tilde{D}_{T,i}$ as the terminal cash flow.

The discount d_i can be calibrated from empirical secondary-market transaction data—for example, from the Setter Capital secondary transaction dataset, which provides observed bid–ask spreads and transaction prices for LP interests as a function of fund age, strategy, and market conditions. This grounds the NAV adjustment in real transaction data rather than arbitrary assumptions.

Advantages. The approach is conceptually simple and uses the same cumulative SDF for all vintages, maintaining a unified pricing framework. The NAV discount directly addresses the appraisal-bias concern by marking the terminal position to secondary-market levels.

Limitations. The discounted NAV is still treated as a terminal cash flow and discounted over the fund’s entire remaining horizon, so the estimator retains sensitivity to the macro state at the dataset endpoint. The

discount function d_i introduces model risk: it must be estimated or calibrated from secondary-market data that may not be representative of the broader PE universe, and it may vary with market conditions in ways that interact with the SDF specification.

6.2 Approach B: NAV-return SDF for all vintages

An alternative strategy abandons the cash-flow pricing restriction entirely and instead prices quarterly NAV returns for *all* vintages—seasoned and unseasoned alike. For each fund i with quarterly NAV observations, define the gross NAV return

$$R_{t,i} = \frac{CF_{t,i} + \text{NAV}_{t,i}}{\text{NAV}_{t-1,i}},$$

and the one-period Euler pricing error $\eta_{t,i}(\theta) = M_{t-1,t}(\theta)R_{t,i} - 1$. The fund-level moment is then a time-series average of these quarterly Euler errors, and the fold moment aggregates across funds and vintages as in Section 4.

Advantages. The approach is internally consistent: the same pricing restriction applies to all vintages, eliminating the need for a seasoned/unseasoned partition or a NAV discount model. Each quarterly Euler error depends only on the one-period SDF, avoiding the exploding-alpha degeneracy entirely. This approach is closest in spirit to Brown et al. [2023] and Ghysels et al. [2025], who construct high-frequency PE returns from NAV-based state-space models and use them for factor analysis and spanning tests.

Limitations. The approach relies on NAV quality for the *entire* sample, including seasoned vintages where realized cash flows are available and arguably preferable. For seasoned vintages, the cumulative CF-SDF restriction uses actual divestitures whose co-movement with public factors is genuine, whereas the NAV-return Euler restriction uses appraisals whose co-movement may be smoothed or biased. The approach therefore trades the unseasoned-vintage dilemma for a different concern: using appraisals where realized payoffs exist.

6.3 Approach C: hybrid cash-flow / normalized-Euler estimator (proposed)

The proposed approach uses the standard cumulative cash-flow pricing error for seasoned vintages and replaces it with a sequence of normalized one-period Euler restrictions on quarterly NAV returns for unseasoned vintages. Crucially, we do *not* discount those quarterly errors back to the initial date with the candidate cumulative SDF, because that would reconstruct the same long-horizon terminal-NAV exposure that Approach A retains. Instead, we normalize each quarterly error by lagged capital at risk, average within fund, and then rescale the result by an exogenous duration factor chosen to make young-fund moments comparable in magnitude to seasoned NPV moments.

One-period Euler pricing error. For a fund i in an unseasoned vintage with quarterly NAV observations at dates $t = 0, 1, \dots, T_i$, define the one-period pricing error for quarter t :

$$\eta_{t,i}(\theta) = \Psi_{t-1,t}(\theta) \cdot (CF_{t,i} + \text{NAV}_{t,i}) - \text{NAV}_{t-1,i},$$

where $CF_{t,i}$ is the net cash flow during the quarter, $\text{NAV}_{t,i}$ is the reported net asset value at quarter-end, and $\Psi_{t-1,t}(\theta)$ is the one-period stochastic discount factor.

The restriction $E[\eta_{t,i}(\theta_0)] = 0$ is an Euler equation: the beginning-of-period NAV equals the risk-adjusted present value of within-period distributions plus the end-of-period NAV.

Normalization and scale matching. Define the lagged capital-at-risk scale

$$s_{t-1,i} = \max\{\text{NAV}_{t-1,i}, \nu K_i\}, \quad \nu \in [0.01, 0.05],$$

and the relative quarterly Euler error

$$r_{t,i}(\theta) = \frac{\eta_{t,i}(\theta)}{s_{t-1,i}}.$$

Because $s_{t-1,i}$ is measurable at date $t-1$, $E[r_{t,i}(\theta_0)] = 0$ whenever $E[\eta_{t,i}(\theta_0)] = 0$. The floor νK_i prevents the denominator from becoming arbitrarily small in the last few quarters of a nearly liquidated fund.

Next average within fund:

$$\bar{r}_i(\theta) = \frac{1}{T_i} \sum_{t=1}^{T_i} r_{t,i}(\theta).$$

This object is a mean quarterly pricing error per dollar at risk. To map $\bar{r}_i(\theta)$ into NPV-like units, let $s(i)$ denote the strategy of fund i , let $\bar{u}_s(a)$ be a benchmark age- a profile of NAV/ K for strategy s , and let q_{a-1} be a deterministic discount sequence (baseline $q_{a-1} = 1$, optional risk-free discounting as a robustness check). Define the duration-matching factor

$$\Lambda_i = c_{s(i)} \sum_{a=1}^{T_i} q_{a-1} \bar{u}_{s(i)}(a),$$

where $c_{s(i)}$ is an optional strategy-specific calibration constant. In the simplest implementation $c_{s(i)} = 1$. If the raw duration scale still leaves systematic size gaps, $c_{s(i)}$ can be chosen so that near-boundary seasoned and unseasoned funds have similar median absolute pricing moments within strategy.

The normalized unseasoned fund moment is

$$g_i^U(\theta) = \Lambda_i \bar{r}_i(\theta),$$

and for an unseasoned vintage $\nu \in \mathcal{V}^U$ and fund cell $C_{n,\nu,r}$, define the cell-level pricing error as

$$\epsilon_{n,\nu,r}^U(\theta) = \frac{1}{|C_{n,\nu,r}|} \sum_{i \in C_{n,\nu,r}} g_i^U(\theta).$$

This normalization has a simple economic interpretation. The term $\bar{r}_i(\theta)$ is the average quarterly pricing error per dollar at risk. The factor Λ_i is an externally chosen capital-at-risk duration: it measures how much NAV/ K exposure a typical fund of that age and strategy carries over its observed life. If the relative quarterly mispricing is roughly stable, $r_{t,i}(\theta) \approx \rho_i(\theta)$, then

$$g_i^U(\theta) \approx \rho_i(\theta) \Lambda_i,$$

so the young-fund moment is of the same order as a life-of-fund NPV error per committed dollar. This is the sense in which the normalization is “smart”: it matches scales without using the candidate cumulative SDF itself.

Advantages. Realized cash flows are used where available (seasoned vintages); NAV-based moments are confined to unseasoned vintages where no alternative exists. The short-horizon Euler restriction avoids both the terminal-NAV discounting of Approach A and the full-sample NAV reliance of Approach B. The duration-matching normalization makes the young-fund block directly comparable to the seasoned NPV block.

Limitations. The approach introduces a seasoned/unseasoned boundary that is itself a design choice (the threshold δ), as well as the duration-matching normalizer Λ_i , which is calibrated outside the SDF. The resampling framework can test sensitivity to these choices by varying δ and the Λ_i specification and comparing holdout losses.

6.4 Letting the framework decide

The three approaches differ in how they treat unseasoned vintages, but all produce vintage-level pricing errors $\epsilon_{n,v,r}(\theta)$ that plug into the same fold-moment structure:

$$m_{n,r}(\theta; M, \eta) = \sum_{v \in A_{n,r}} \omega_{n,v,r} \epsilon_{n,v,r}(\theta; M, \eta).$$

The entire estimation and validation machinery of Section 5 applies without modification to each approach individually. Within a given approach, the resampling-based holdout loss provides a clear ranking of SDF models: the model with the lowest average holdout pricing loss is the one that best prices held-out moments *as defined by that approach*.

Cross-approach comparability. Comparing holdout losses *across* approaches is less straightforward, because the three approaches define different moment conditions. Approach A evaluates a cumulative CF-SDF error with an adjusted terminal NAV; Approach B evaluates one-period Euler errors on NAV returns; Approach C evaluates a mixture of cumulative CF errors and normalized Euler errors. These moment conditions differ in scale, economic content, and sensitivity to the SDF parameters, so a holdout loss of, say, 0.02 under one approach and 0.05 under another does not imply that the first approach “works better”—the two numbers measure different things.

This is an important open question that requires careful resolution. One possible path is to separate *estimation* from *evaluation*: estimate $\hat{\theta}$ using each approach’s own moments, but evaluate all three on a *common* criterion that does not depend on the approach. A natural candidate is the seasoned-vintage cash-flow pricing error, since realized cash flows are the gold standard and all three approaches agree on their treatment of seasoned vintages. Each approach produces a different $\hat{\theta}_n^{(k)}$, but one can always evaluate

$$S_n^{\text{common}}(\hat{\theta}_n^{(k)}) = L \left(\sum_{v \in \mathcal{V}^S \cap H_n} \omega_{n,v,H_n} \epsilon_{n,v,H_n}^S(\hat{\theta}_n^{(k)}) \right), \quad k \in \{A, B, C\},$$

asking: *which estimation approach produces the $\hat{\theta}$ that best prices held-out realized cash flows from seasoned vintages?* A complementary strategy is forward-looking (out-of-time) validation: estimate the SDF on the early part of the panel—where then-unseasoned vintages are treated differently by each approach—and evaluate on now-seasoned vintages using realized cash flows. Both strategies remain to be developed and tested in the empirical implementation.

Diagnostics. To diagnose whether the estimator disproportionately fits one group, the empirical implementation should report separate holdout losses for $\mathcal{V}^S$ and $\mathcal{V}^U$:

$$S_n^S(M, \eta, \Pi) = L \left(\sum_{v \in \mathcal{V}^S \cap H_n} \omega_{n,v,H_n} \epsilon_{n,v,H_n}^S(\hat{\theta}_n) \right), \quad S_n^U(M, \eta, \Pi) = L \left(\sum_{v \in \mathcal{V}^U \cap H_n} \omega_{n,v,H_n} \epsilon_{n,v,H_n}^U(\hat{\theta}_n) \right).$$

If the average $\bar{S}^U$ is systematically lower than $\bar{S}^S$, the unseasoned-vintage restrictions are being trivially satisfied and contribute little to identification; if $\bar{S}^U \gg \bar{S}^S$, the short NAV histories are dominating the holdout loss and the model selection may be driven by NAV fit rather than cash-flow fit.

7 Identification and Stability

The exploding-alpha issue—where an arbitrarily large α drives cumulative discount factors to zero, creating a ridge of spurious near-minima—is a *finite-sample* pathology, not an asymptotic one. Under increasing-domain asymptotics indexed by vintage years, as $V \rightarrow \infty$ the cross-vintage moment averages over many independent realizations of the cumulative SDF. The global minimum of the population criterion $Q(\theta)$ is uniquely attained at θ_0 under standard identification conditions, and the spurious near-minima created by extreme alpha values vanish: no single alpha value can simultaneously zero out the discounted cash flows of many independent vintage realizations with heterogeneous timing and factor paths.

In finite samples, the exploding-alpha ridge is controlled by three complementary mechanisms:

1. **Short-horizon Euler restrictions.** Under the hybrid approach C, unseasoned vintages contribute one-period Euler errors whose sensitivity to α is proportional to the quarter length Δt , not exponential in the fund horizon T . The lagged scale $s_{t-1,i}$ and duration-matching factor Λ_i are exogenous to the candidate cumulative SDF and do not reintroduce long-horizon alpha sensitivity. Under the full NAV-return approach B, all vintages use one-period discounting, eliminating long-horizon alpha sensitivity entirely. Even under approach A, the NAV discount reduces the terminal position, partially mitigating the problem.
2. **Resampling regularization.** The validation fold penalizes degenerate alpha values because extreme parameter estimates that overfit the estimation folds will produce large holdout pricing errors. Across many resampled splits, degenerate solutions are revealed by high holdout-loss variance and frequent boundary solutions.
3. **Asymptotic cash-flow dominance.** Under approaches A and C, only the most recent vintages within a bounded horizon D are classified as unseasoned. As the panel grows, $|\mathcal{V}^U|/V \rightarrow 0$, so the contribution of any NAV-based block to the cross-vintage average vanishes. The estimator is therefore asymptotically cash-flow dominated.

Together, these mechanisms ensure that the exploding-alpha issue is a finite-sample nuisance controlled by the estimation design, not a fundamental limitation of the SDF pricing framework.

NAV quality and the Euler moment condition. The Euler moment condition $E[\eta_{t,i}(\theta_0)] = 0$ makes a stronger claim: that the beginning-of-period NAV equals the risk-adjusted present value of within-period cash flows plus the end-of-period NAV. For traded assets, no-arbitrage forces this condition to hold. For private-equity funds, NAVs are appraisals whose co-movement with public factors may be smoothed or biased, so the Euler restriction holds only approximately. The paper’s design controls this concern in three ways. First, under approaches A and C, only unseasoned vintages rely on NAV-based moments; seasoned vintages are priced with realized cash flows alone. Second, even within $\mathcal{V}^U$, the normalized Euler block uses one-quarter NAV changes and lagged scales rather than a single large terminal NAV discounted over many years. Third, the asymptotic irrelevance property ensures that the fraction of moment conditions relying on NAV quality vanishes as the panel grows. Under approach B, NAV quality affects the entire sample, which is the price paid for internal consistency.

8 Alpha Design

The resampling method does not by itself decide what alpha should mean. The cleanest baseline is to use a cumulative, NPV-style alpha rather than a period-by-period intercept alpha.

8.1 Baseline: cumulative alpha

For fold $F_{n,j}$, let $B_t(\beta)$ be the cumulative benchmark return implied by factor loadings β . A direct-alpha-style fold moment is

$$\psi_{n,j}(a, \beta) = \sum_t \frac{c f_{n,j,t}}{B_t(\beta)} \exp(-a \tau_{n,j,t}),$$

where $c f_{n,j,t}$ is the fold cash flow at time t , and $\tau_{n,j,t}$ is elapsed time from the fold origin to date t . For fixed β , the fold-implied cumulative alpha solves

$$\sum_t \frac{c f_{n,j,t}}{B_t(\beta)} \exp(-a_{n,j}(\beta) \tau_{n,j,t}) = 0.$$

This is close to the KN16/GPME/direct-alpha tradition because alpha is an NPV-style abnormal-performance object [Korteweg and Nagel, 2016, Gredil et al., 2023, Korteweg and Nagel, 2025]. It is also more natural for long-horizon PE cash flows than a monthly intercept alpha. The exponential $\exp(-a \tau)$ parameterizes alpha as a continuously compounded rate; for estimation this is equivalent to the discretely compounded $(1 + \alpha)^{-t}$ up to the reparameterization $a = \log(1 + \alpha)$.

The fold-implied alpha $a_{n,j}(\beta)$ serves a dual purpose in the resampling design. In *estimation*, a common alpha a is estimated jointly with β across the P estimation folds by minimizing the aggregate loss over fold moments $\psi_{n,j}(a, \beta)$. In *diagnostics*, the fold-specific implied alpha $a_{n,j}(\hat{\beta})$ is computed for each fold (including the validation fold) by solving $\psi_{n,j}(a, \hat{\beta}) = 0$ at the estimated $\hat{\beta}$. The cross-fold dispersion of $\{a_{n,j}\}_{j=1}^{P+h}$ is then a model diagnostic: a large spread indicates that the alpha estimate is sensitive to which funds enter the fold, and therefore that the aggregate alpha is weakly identified.

8.2 Alternative alpha definitions

The resampling framework can also compare alpha definitions as part of the design vector η : (i) the cumulative alpha baseline above, (ii) a DLP12-style intercept alpha, useful for comparison but vulnerable to exploding-alpha behavior in small or weakly identified samples [Driessen et al., 2012, 2008], and (iii) an error-term alpha ($\mathbb{E}[\epsilon]$) as in the machine-learning approach of Tausch and Pietz [2024]. The empirical question is not only which SDF factors perform best, but also which alpha definition yields stable and predictive PE pricing moments.

9 Vintage-Stratified Factor Models

The resampling framework naturally accommodates time-varying factor exposures. Instead of estimating a single β across all vintages, define vintage-bucket-specific loadings β_b for vintage bucket $b \in \{B_1, \dots, B_K\}$, where each bucket B_k collects vintages from a contiguous window (e.g., 2000–2004 buyout, 2015–2019 venture capital). The parameter vector becomes $\theta = (\alpha, \beta_{B_1}, \dots, \beta_{B_K})'$, and the resampling framework applies

without modification: more parameters require more estimation folds, and the holdout loss automatically penalizes overfitting.

Floating buckets. Overlapping vintage windows centered on each year—for example, a five-year window centered on 2010 that includes vintages 2008–2012—provide smooth estimates of how factor exposures evolve over time without sharp boundary effects. The bucket width is itself a tunable parameter that the resampling holdout loss can optimize.

Strategy-specific buckets. Factor exposures likely differ across PE strategies. A natural extension defines buckets jointly over vintage year and strategy (e.g., “2002 buyout factor,” “2020 venture capital factor”), so that the holdout loss can compare: (i) pooled- β vs. bucket- β , (ii) narrow vs. wide bucket windows, and (iii) strategy-specific vs. pooled-strategy buckets.

This vintage-stratified specification showcases the flexibility of the resampling framework. The question of whether factor exposures are time-varying is empirically testable within the framework, not assumed a priori. A model with vintage-stratified betas that achieves lower holdout loss than a pooled-beta model provides direct evidence of time variation in PE factor exposures—a finding of independent interest for the asset-pricing literature.

10 Literature Positioning

The paper can be positioned relative to the existing PE SDF literature along four dimensions:

- **From DLP12:** price private cash flows directly with an SDF [Driessen et al., 2012].
- **From KN16:** use aggregated pricing moments and an NPV-style abnormal-performance object [Korteweg and Nagel, 2016, 2025].
- **New here:** a general resampling-based model selection framework that treats the construction of PE pricing moments as a testable design choice. Instead of committing to one grouping, alpha definition, or unseasoned-vintage treatment, the framework varies these choices systematically and ranks them by holdout pricing loss.
- **Also new:** a three-competitor empirical horse race for unseasoned vintages—NAV-adjusted CF, full NAV returns, and hybrid normalized Euler—evaluated within a common resampling framework. The framework can further accommodate vintage-stratified factor models to test for time-varying risk exposures.

The paper is therefore not a repair of any existing SDF approach. It is a new empirical-design paper: how should we build, resample, and validate PE pricing moments when the sample is small and the moment construction is consequential? The resampling framework is the hero; individual design choices—moment type, alpha definition, unseasoned-vintage treatment, vintage-stratified betas—are its inputs.

11 Empirical Algorithm

For each candidate model M , alpha design η , unseasoned-vintage approach (A, B, or C), and partition scheme Π :

1. Partition vintages into seasoned ($\mathcal{V}^S$) and unseasoned ($\mathcal{V}^U$) sets (if using approach A or C).
2. Choose P estimation folds and h holdout folds.

3. Draw a sampled assignment n from Π .
4. Build fold cash-flow portfolios and fold pricing moments $m_{n,r}(\theta; M, \eta)$.
5. Estimate $\hat{\theta}_n(M, \eta, \Pi)$ on the P estimation folds.
6. Evaluate the fitted model on the holdout fold.
7. Repeat over many sampled assignments.
8. Rank model-design pairs by average holdout pricing loss.

Alongside the mean holdout loss, report diagnostics that help interpret the ranking:

- median and dispersion of holdout losses,
- dispersion of $\hat{\theta}_n$ across assignments,
- frequency of non-convergence,
- frequency of boundary or exploding-alpha solutions,
- separate holdout losses for $\mathcal{V}^S$ and $\mathcal{V}^U$,
- sensitivity of results to within-vintage resampling versus shuffled vintage-year portfolios.

A Detailed Comparison of Pricing Moments

This appendix provides a detailed comparison of the dollar NPV and PME-ratio moment families. The main text uses the NPV moment as the baseline; the PME-ratio serves as a robustness check.

Dollar NPV moment. For seasoned funds in fold cell j in split n , aggregate discounted cash flows into one cumulative pricing error,

$$m_{n,j}^{\text{NPV}}(\theta; M, \eta) = \sum_{i \in C_{n,j}} w_i \sum_{t=0}^{T_i} \Psi_{0,t}(\theta; M, \eta) CF_{t,i}, \quad CF_{t,i} = D_{t,i} - X_{t,i},$$

where $X_{t,i}$ are contributions, $D_{t,i}$ distributions (including terminal NAV for seasoned funds), and $\Psi_{0,t}$ the cumulative SDF. At the population parameter, $E[m_{n,j}^{\text{NPV}}(\theta_0)] = 0$: the risk-adjusted NPV of fund cash flows vanishes.

Pros. (i) *Linearity* in cash flows yields tractable GMM derivatives, influence functions, and CLT-based inference. (ii) *Additivity* across funds makes the fold structure of Section 11, as well as subsampling and the bootstrap, transparent: the cell moment is the sum of fund-level moments, and the split moment is the sum of cell moments. (iii) *Compatibility with the normalized Euler NAV restriction* for unseasoned vintages: the seasoned block remains a linear cash-flow NPV, while the unseasoned block uses lagged-scale one-period Euler errors rescaled by an exogenous duration factor. Both blocks are expressed as per-committed-dollar pricing errors, and neither requires ratio denominators that can approach zero.

Cons. (i) *Scale-dependence.* A \$1B fund with a 1% relative NPV error contributes ten times more to the moment than a \$100M fund with the same relative error, so efficient GMM weighting implicitly tilts the estimator toward mega-funds. The influence function is mechanically heavy-tailed in the fund-size distribution rather than in genuine pricing disagreement. (ii) *Dollar-denominated comparisons across*

vintages with heterogeneous aggregate commitments are not directly meaningful; normalization is left to the econometrician. (iii) *Sign cancellation*. A cross-sectional mean of zero is consistent with systematic underpricing of one subgroup offsetting overpricing of another, which the scalar moment cannot detect. This motivates the fine-grained fold structure but does not eliminate the concern within a fold.

PME-ratio moment. Define the risk-adjusted PME of fund i as

$$\text{PME}_i(\theta; M, \eta) = \frac{\sum_t \Psi_{0,t}(\theta; M, \eta) D_{t,i}}{\sum_t \Psi_{0,t}(\theta; M, \eta) X_{t,i}},$$

from which two natural cell-level moments follow: the *average-PME* form $m_{n,j}^{\text{PME}}(\theta) = |C_{n,j}|^{-1} \sum_{i \in C_{n,j}} (\text{PME}_i(\theta) - 1)$ and the *pooled-PME* form $m_{n,j}^{\text{PME-pool}}(\theta) = (\sum_{i,t} \Psi_{0,t} D_{t,i}) / (\sum_{i,t} \Psi_{0,t} X_{t,i}) - 1$. Both have mean zero at θ_0 under standard regularity, but they estimate different population objects (mean of a ratio versus ratio of means).

Pros. (i) *Scale-invariance*. The ratio is unit-free, so fund-size heterogeneity does not mechanically up-weight mega-funds. A \$100M fund and a \$1B fund enter the cross-sectional moment on equal footing. (ii) *Practitioner interpretability*. PME is the native performance metric of the industry; reporting results in the same units aids dissemination. (iii) *Bounded downside*. The fund-level PME is bounded below by zero, so $\text{PME}_i - 1$ is bounded below by -1 : a single disaster fund cannot dominate the moment the way it can dominate a dollar NPV.

Cons. (i) *Nonlinearity*. The moment is a ratio in θ ; its derivative involves the ratio of two correlated sums and is numerically fragile when the denominator (discounted contributions) is small. This occurs mechanically for young funds with back-loaded capital calls — precisely the vintages that the Euler NAV restriction is designed to retain. (ii) *Jensen-style bias between forms*. The average-PME and pooled-PME moments can differ in finite samples by a term of order $\text{Var}(X_i)/\bar{X}^2$; the two are asymptotically equivalent only under degenerate size distributions. Researchers must commit to one and defend it. (iii) *No clean short-horizon analogue for unseasoned vintages*. Integrating unseasoned vintages $\mathcal{V}^U$ into a PME framework typically requires treating the latest $\text{NAV}_{T_i,i}$ as a terminal distribution and discounting it over the fund’s entire remaining horizon — exactly the long-horizon discounting mechanism that the normalized Euler block is designed to avoid. (iv) *Inference*. Upper-tail skewness in PME_i (funds can return many times their capital, but cannot lose more than it) distorts CLT-based standard errors and t -statistics unless explicitly robustified, e.g., by bootstrapping the cell-level moment or by employing median-based loss functions.

B Why We Do Not Discount Euler Errors to Time Zero

Multiplying $\eta_{t,i}(\theta)$ by $\Psi_{0,t-1}(\theta)$ and summing would yield a valid moment, and for fully liquidated funds it would telescope. But for the young funds of actual interest it would recreate the long-horizon terminal-NAV term:

$$\sum_{t=1}^{T_i} \Psi_{0,t-1}(\theta) \eta_{t,i}(\theta) = \sum_{t=1}^{T_i} \Psi_{0,t}(\theta) CF_{t,i} + \Psi_{0,T_i}(\theta) \text{NAV}_{T_i,i} - \text{NAV}_{0,i}.$$

For an unseasoned fund, $\text{NAV}_{T_i,i} \neq 0$, so discounting the quarterly Euler errors back to time zero reconstructs exactly the large terminal-NAV present-value term that generates instability. The normalized construction in Section 6.3 sacrifices path-by-path telescoping in order to preserve the short-horizon dependence of the young-fund block on the candidate SDF.

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